

DG Lie Algebroid and Quotient Stack

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In this paper, I'll discuss foliation from point of view of derived algebraic geometry. Derived algebraic geometry proves the ∞ -equivalence of dg Lie algebroid and foliation. Now the quotient is defined by quotient of dg L_∞ groupoid generated from dg Lie algebroid, and it's calculated by dg L_∞ version of Chevalley-Eilenberg complex. I'm also wondering evaluation of the foliation using language of DT/GW invariants. First of all, the idea of foliation is classically defined differential geometrically on a smooth manifold:

Definition 0.1. Foliation

Foliation is a partition of a smooth manifold M into its submanifolds. One idea of generating foliation is to retrieve from distribution $D_x \subset TM$ for some $x \in M$, and this distribution classifies the points $x \in M$ to define foliation. Distribution D_x is defined by its vector field $X, Y \in \Gamma(D_x)$ satisfying $[X, Y] \in \Gamma(D_x)$, and in this case D_x is called integrable. This D_x corresponds to a submanifold L , since there's its restriction $D_x|_L = T_x L$ by the unique maximal submanifold $L \subset M$ s.t. $x \in L$. Note in fact $D_x|_{L_y} = T_y L$ for all $y \in L$. Accordingly, the resulting set $\{L_y\}$ for $y \in M$ is the set of submanifolds $L_y \subset M$ and disjoint union $\coprod L_y = M$.

To verbalize the notion of foliation is an equivalent notion with integrability by Frobenius theorem, where integrability is done by Lie algebra. If Lie algebra describes the global property (there's no "if" in math Toynbee says), the corresponding Lie group quotient of manifold $[M/G]$ is a stack. It doesn't work, however. I'll use Lie algebroid to describe every point of the submanifold $L \subset M$, whose Lie groupoid $\mathcal{G}$ quotient is a quotient stack, which describes foliation $[M/\mathcal{G}]$. The first step of foliation my plan is to modernize the definition of foliation by using quotient stack with Lie algebroid/groupoid.

Definition 0.2. Groupoid

A groupoid is a category in which every morphisms are invertible. Denote the objects are $Ob(\mathcal{C}) = G_0$, and the morphisms are $Mor(\mathcal{C}) = G_1$, the double arrow $G_1 \rightrightarrows G_0$. For an arrow $g : x \rightarrow y$, there are source and target maps $s, t : G_1 \rightarrow G_0$, together with composition $G_1 \times_{G_0} G_1 \rightarrow G_1$, identity arrow and inverses. A group becomes a Lie groupoid if G_0 and G_1 are smooth manifolds, and all the maps are smooth.

There are many important examples of Lie groupoid. The trivial things (logically equivalent to Lie group but somehow interesting generalization) are Lie group, which is a Lie groupoid with one object so $G \rightrightarrows *$. Another example which is more geometric is that Lie group action $G \times X \rightrightarrows X$ can be modeled by $G_1 = G \times X$ and $G_0 = X$. By the context of foliation, Lie groupoid $Hol(\mathcal{F})$ of the foliation $\mathcal{F}$ is holonomy groupoid. All the above examples are important. Now we consider quotient stack by Lie groupoid, in particular by Lie group. The definition of quotient stack below is a moduli stack while keeping the group symmetry structure for group G , but groupoid quotient $\mathcal{G}$ is somewhat more complicated:

Definition 0.3. Quotient Stack

For an algebraic variety X , quotient stack by Lie group G is $[X/G]$ is a functor $[X/G] : (Spaces)^{op} \rightarrow Groupoids$.

$$[X/G](S) = \{(P \rightarrow S, f : P \rightarrow X) | P \rightarrow S \text{ is a principal } G\text{-bundle, } f_*f^* = \text{id}_X\}$$

In particular if $X = pt$, the quotient stack is $[pt/G] = BG$, and $BG(pt)$ is one-object groupoid whose automorphism group is $G = Hom(*, *)$.

Meanwhile, quotient stack by groupoid $[U/R]$ for a groupoid R is a stackification $[U/R] = St([U/_p R])$ of category fibered in groupoids $p : U \rightarrow R$. This works for all $c \in R$, there exists $x \in U$ s.t. $p(x) = c$, and for all $f : x \rightarrow y$, $p(f) = id_c$.

where I abbreviate what the stackification means in this introductory chapter for the moment. In any case, the important quotient stack is holonomy groupoid quotient $[M/Hol(F)]$, which is a foliation. But how does Lie groupoid appears, where Lie algebra solves integrability condition in manifold? Lie algebra can be generalised to Lie algebroid by its globalization, where Lie groupoid comes. The definition of Lie algebroid is:

Definition 0.4. Lie Algebroid

For a smooth manifold M , a Lie algebroid over M consists of $(A \rightarrow M, [\cdot, \cdot]_A, \rho)$ where $A \rightarrow M$ is a vector bundle, $[\cdot, \cdot]_A : \Gamma(A) \times \Gamma(A) \rightarrow \Gamma(A)$ is a Lie bracket of sections, $\rho : A \rightarrow TM$ is an anchor map. These satisfy product rule $[X, fY]_A = f[X, Y]_A + \rho(X)(f)Y$, and the anchor preserves bracket $\rho([X, Y]_A) = [\rho(X), \rho(Y)]$.

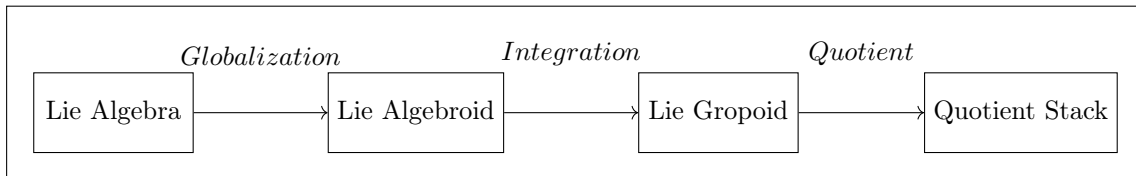
This Lie algebroid is a generalization of Lie algebra. In fact, the fiber A_x $x \in M$ of vector bundle $A \rightarrow M$ is Lie algebra, so containing the data of local symmetry.

Remark 0.1

As a comparison, quotient stack $[M/G]$ of gauge theory is a Lie group quotient, not Lie groupoid, where the local symmetry of particle action is everywhere the same, but the construction is nothing to do with the foliation problem. In other words, not all quotient stack is foliation.

The computation of quotient stack $[M/G]$ is done by Chevalley-Eilenberg complex. There's generalization of Chevalley-Eilenberg for the case of quotient stack by Lie groupoid.

In summary, the above argument can be depicted in the following flow chart, and now that the foliation is modernized geometrically, studying foliation is somewhat equivalent to studying quotient stack:



Several questions appear from this informal writing of foliation. First of all, to make the argument more precisely, can Lie algebroid be always integrated to the Lie groupoid? Now true in general, but Crainic-Fernandes integrability criterion claims Lie algebroid used in foliation can be always integrated.

Proposition 0.1. Crainic-Fernandes Criterion

TBD

Now let's set some research questions. First of all, to introduce the application regarding questions of foliation beyond of itself, QFT might be the obvious application, while we also further investigate the property of quotient stack itself. Perhaps subjectively, Lie groupoid quotient is much more dirty than that of Lie group, but in order to make it something, considering the M as simple as possible, hyperkähler manifold, for example. I'd also

consider the computational method with Chevalley-Eilenberg, Donaldson-Thomas/Gromov-Witten invariants. Another problem is to discuss what's the missing data when we define quotient stack of Lie groupoid.

Question 0.1

(application)

- What happens if the foliation is Lagrangian?
- Is there any connection to Fukaya category?
- Is there any either deformation or quantization?

(structure)

- Restricting geometry M to hyperkähler, or as simple as possible?
- Computation of groupoid version of CE cohomology? DT/GW invariants?
- Does quotient stack contains all the necessary information?

1 Algebraic Geometry

According to toën's paper.

1.1 Classic Algebraic Geometry

GIT

1.2 Derived Algebraic Geometry

Definition 1.1

Definition 1.2

2 QFT World

Definition 2.1. Lagrangian

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